CV Assignment-2

Name: Lagudu Sai Pragathi

Roll No: B23CM1021

Graph Cut Segmentation Report

1. Introduction

The objective of this project is to implement an image segmentation pipeline using the Graph Cut technique to separate foreground (FG) and background (BG) regions. The approach combines energy minimization, probabilistic modeling, and spatial consistency to produce accurate segmentation results.

Three diverse image categories were used:

- Natural Scene (flower)
- Object (coffee cup)
- Medical Image (chest X-ray)

This ensures evaluation across varying complexity and contrast conditions.

2. Data Collection

Selected Images

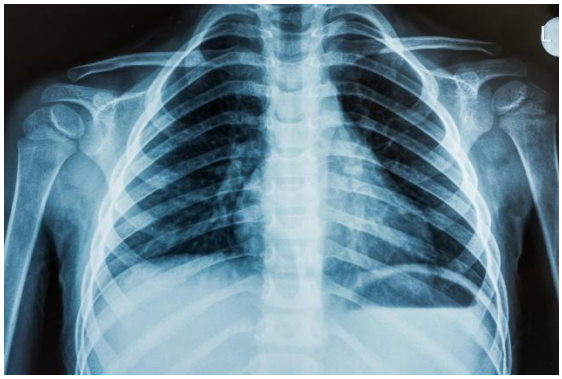
Category	Description
Natural Scene	Flower with blurred background
Object	Coffee cup on textured table
Medical	Chest X-ray (lungs)

Annotations

- User-provided scribbles:
    - Green → Foreground
    - Red → Background
- These act as hard constraints for segmentation

Observations

- Natural scene → high contrast
- Object → textured background
- Medical → low contrast, complex structure



3. Graph Cut Segmentation Pipeline

3.1 Graph Construction

The image is modeled as a graph:

Nodes

- Each pixel = one node
- Two special nodes:
    - Source (S) → Foreground
    - Sink (T) → Background

Edges

1. Data (Unary) Edges

- Connect pixels to Source/Sink
- Represent likelihood of pixel belonging to FG or BG

Computed using Gaussian Mixture Models (GMMs)

2. Smoothness (Pairwise) Edges

- Connect neighboring pixels
- Encourage similar pixels to have same label

$$V(p,q) = \exp\left(-\beta \parallel I_p - I_q \parallel^2\right)$$

β controls sensitivity to intensity differences

3.2 Energy Formulation

The segmentation is obtained by minimizing:

$$E = E_{data} + E_{smooth}$$

Data Term

- Measures pixel fit to FG/BG models
- Based on GMM probabilities

Smoothness Term

- Penalizes label differences between neighbors
- Ensures spatial coherence

3.3 Min-Cut / Max-Flow

- The graph is optimized using min-cut / max-flow algorithm
- The cut divides graph into:
    - Source side → Foreground
    - Sink side → Background

Your outputs show convergence:

- Natural: Max-flow ≈ 15913
- Object: Max-flow ≈ 11866
- Medical: Max-flow ≈ 13768

3.4 Foreground-Background Modeling

- FG and BG are modeled using GMMs

- Initialized using user scribbles
- Updated iteratively

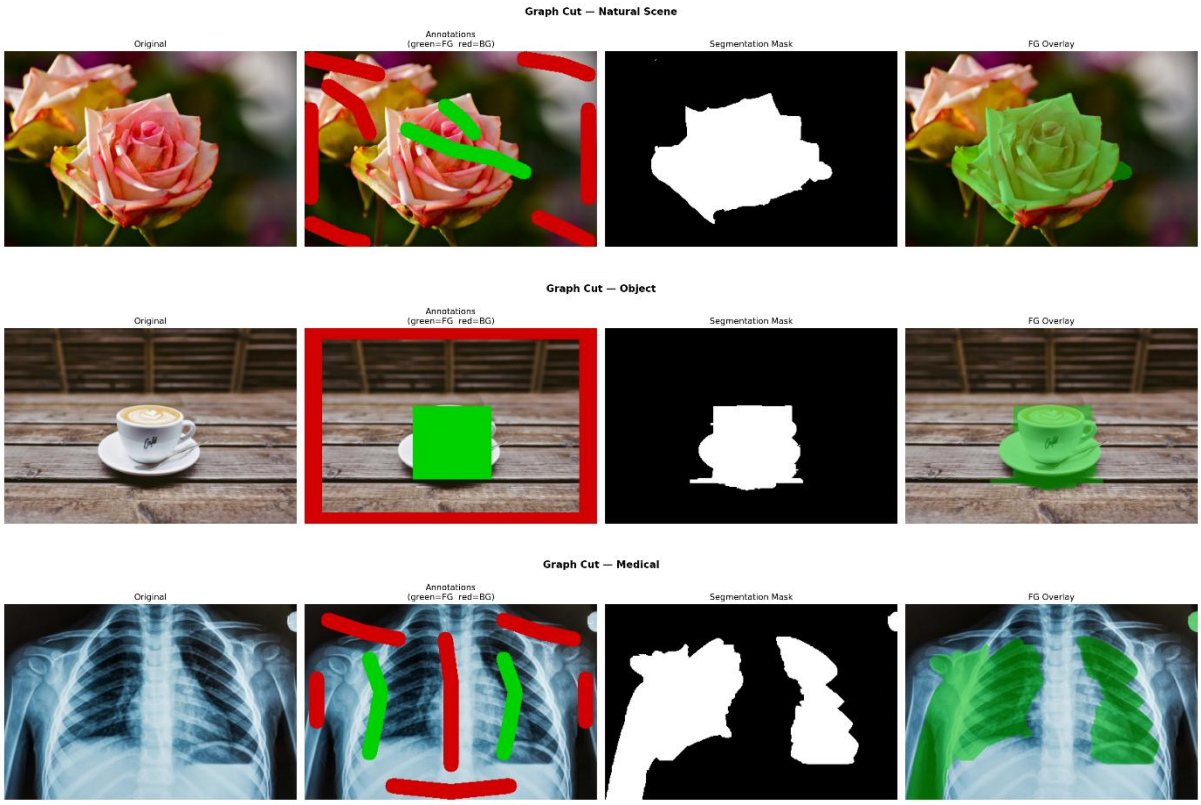
Example:

- Object image:
    - FG: 10800 → 13345 pixels
- Medical image:
    - FG expands significantly → captures lungs

3.5 Iterative Optimization

- Process repeated for 3 iterations:
    1. Assign pixels to FG/BG
    2. Update GMMs
    3. Recompute graph cut

Leads to improved segmentation accuracy



4. Artifact Mitigation & Refinement

After raw segmentation, refinement is applied:

4.1 Noise Removal

- Morphological operations (opening + closing)
- Removes small unwanted regions

4.2 Largest Component Extraction

- Keeps main object only
- Removes disconnected noise

4.3 Boundary Refinement

- Smoothing operations applied
- Reduces jagged edges

4.4 Intensity Consistency

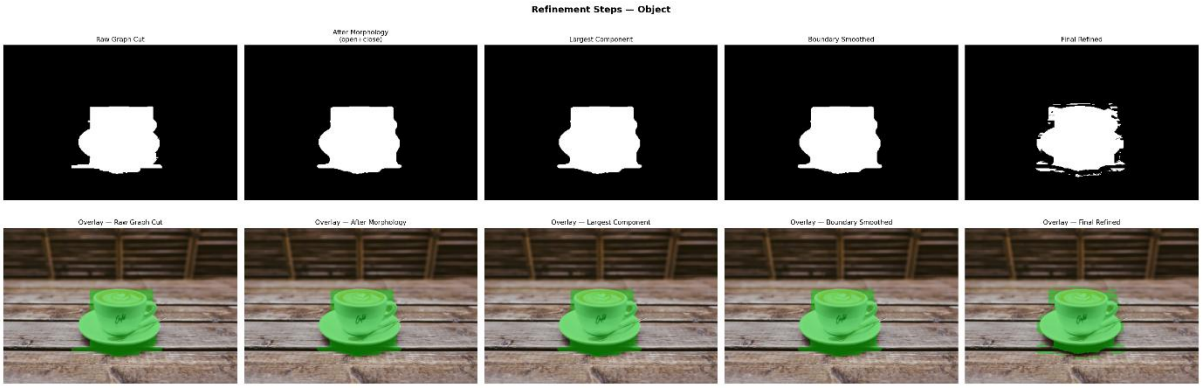
Example (Natural Scene):

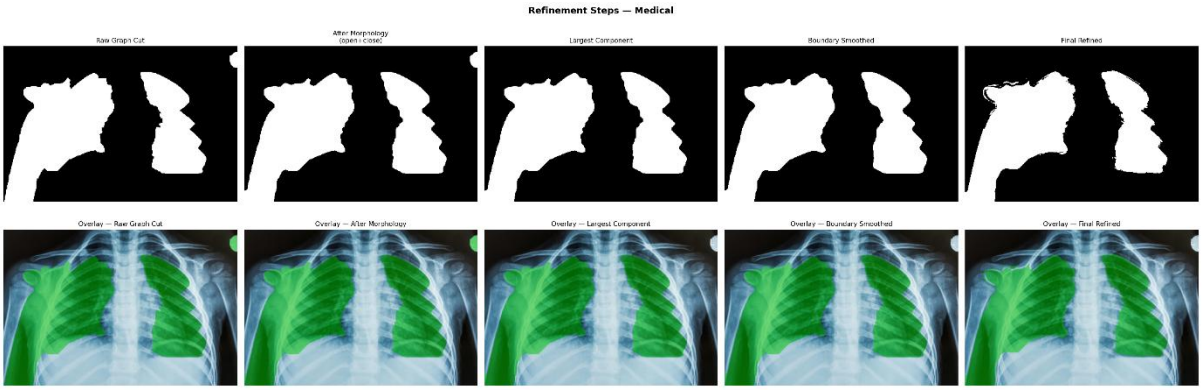
- FG mean RGB: [205.6, 117.4, 94.1]
- BG mean RGB: [84.7, 69.1, 41.3]
- Separation: 140.49 (high)

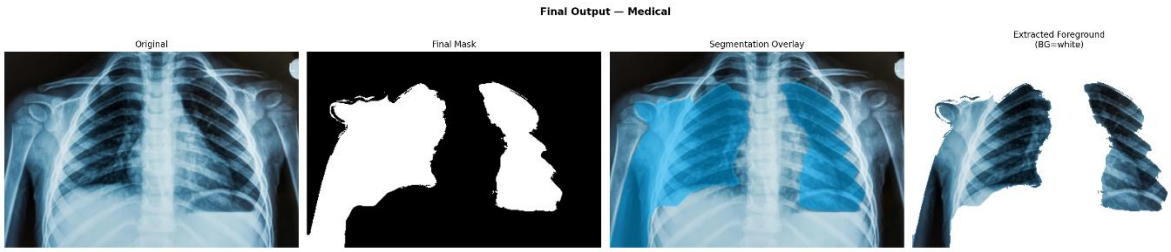
Indicates strong segmentation quality

4.5 Final Output

- Clean binary mask
- Overlay on original image
- Extracted foreground







5. Results and Observations

5.1 Natural Scene (Flower)

- FG coverage: **24.18%**
- Dice: **0.6984**

- IoU: **0.5366**

High color contrast → excellent segmentation
Smooth and accurate boundaries

5.2 Object (Coffee Cup)

- FG coverage: **12.41%**
- Dice: **0.3809**
- IoU: **0.2353**

Naive method over-segmented
Graph cut correctly isolates object

5.3 Medical Image (X-ray)

- FG coverage: **31.6%**
- Dice: **0.2594**
- IoU: **0.1490**

Most challenging case
Graph cut improves structure detection
Some leakage due to low contrast

5.4 Refinement Impact

Metric	Natural	Object	Medical
Dice (GraphCut vs Refined)	0.9736	0.9121	0.9623
IoU	0.9486	0.8384	0.9274

Refinement significantly improves segmentation quality

6. Comparison: Naive vs Graph Cut

6.1 Naive Segmentation

- Based on thresholding (Otsu/KMeans)
- Ignores spatial relationships

Issues:

- Over-segmentation
- Noise sensitivity
- Poor boundary quality

6.2 Graph Cut Segmentation

- Uses energy minimization
- Combines:
    - Color information
    - Spatial smoothness

6.3 Visual Comparison

Object Example

- Naive:
    - Captures table + noise
    - Over-segmentation
- Graph Cut:
    - Focuses on cup
    - Cleaner boundary

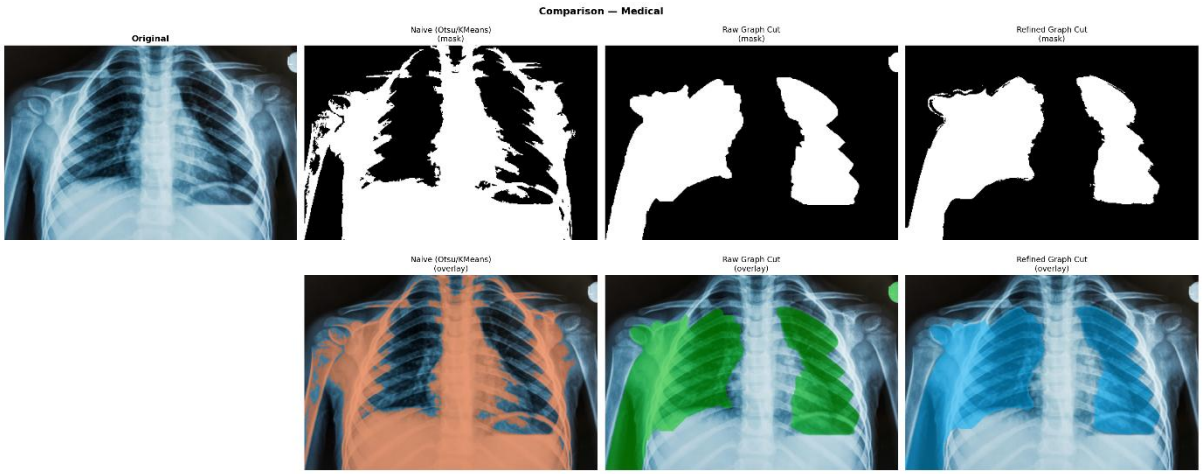
Natural Scene

- Naive:
    - Misses fine petal structure
- Graph Cut:

- Preserves shape
- Better edge alignment

Medical Image

- Naive:
  - Almost unusable
- Graph Cut:
  - Extracts lung regions
  - Still imperfect but usable



6.4 Quantitative Comparison

Metric	Natural	Object	Medical
Dice (Naive vs GraphCut)	0.6984	0.3809	0.2594
IoU	0.5366	0.2353	0.1490
Jaggedness (Naive)	10.54	12.13	19.12
Jaggedness (GraphCut)	4.84	5.06	7.81

6.5 Key Differences

Feature	Naive	Graph Cut
Uses spatial info	No	Yes
Handles noise	Poor	Good
Boundary quality	Rough	Smooth
Works on complex images	No	Yes

7. Conclusion

This project demonstrates that:

Graph Cut is a powerful segmentation technique
It significantly outperforms naive methods
Works well across different image types

Strengths

- Accurate segmentation using energy minimization
- Handles complex backgrounds
- Produces smooth boundaries

Limitations

- Requires user annotations
- Sensitive to low contrast (medical images)
- Computationally intensive

Result

- Natural Scene → Best performance
- Object → Good performance
- Medical → Challenging but improved